

## 2023 AMC 10A Problem 7 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 7. Great practice for AMC 10, AMC 12, AIME, and other math contests

### 2023 AMC 10A Problem 7

**Problem:**

Janet rolls a standard 6-sided die 4 times and keeps a running total of the numbers she rolls. What is the probability that at some point, her running total will equal 3?

**Answer Choices:**

- A.  $\frac{2}{9}$
- B.  $\frac{49}{216}$
- C.  $\frac{25}{108}$
- D.  $\frac{17}{72}$
- E.  $\frac{13}{54}$

**Solution:**

We divide this problem into 4 cases, depending on how Janet gets to the 3:

Janet rolls a total of 3 if she starts by rolling 111, 12, 21, or 3. These initial sequences of rolls occur with probability  $\frac{1}{6^3}$ ,  $\frac{1}{6^2}$ ,  $\frac{1}{6^2}$ , and  $\frac{1}{6}$ , respectively. Adding these probabilities gives

$$\frac{1}{6^3} + \frac{1}{6^2} + \frac{1}{6^2} + \frac{1}{6} = (\text{B}) \boxed{\frac{49}{216}}$$

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